

Worksheet - 7 HOTS Questions

Section A: Counting & Visualization

1. 3-distinct lines intersect exactly at one single point O. How many distinct straight angles are formed at the intersection?
2. You mark four distinct points on paper such that no three points lie on the same straight line. What is the maximum number of distinct line segments you can draw by joining these points?
3. Draw three rays OA, OB & OC originating from the same vertex O. How many distinct angles are formed in this figure?
4. A laser beam is shot from point A, bounces off a mirror at point B, and hits a wall at point C. Name the angle formed at the mirror & identify its vertex.
5. A clock shows 12:00, the minute hand makes one and a half full turns. Where is the minute hand pointing now and what total degree has it rotated?

6. Fold a square piece of paper perfectly along its diagonal. Unfold it. What is the angle between the diagonal crease & the edge of the square paper?

7. Can two acute angles add up to form a straight angle? Explain your reasoning with an example.

8. If you have a protractor that is broken & only has markings from 20° to 150° , can you still use it to measure an angle of 45° ? How?

9. Draw a horizontal line segment AB . Mark a point P on it. Draw a ray PC such that angle APC is an obtuse angle. What type of angle will angle BPC automatically be?

10. You start facing North. You turn 90° clockwise, then 180° anti-clockwise, then 45° clockwise. Which direction are you facing now?